

THE ℓ^1 -BOUNDEDNESS CONJECTURES OF HEIL AND YU

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ABSTRACT. Heil and Yu closed their study of ℓ^1 -bounded sets with two conjectures: that boundedness together with uniform separation does not imply ℓ^1 -boundedness, and that the ℓ^1 -bounded sets are not closed under finite unions. Both are proved here, and a single construction settles them simultaneously: the union of the standard and Fourier bases of $\mathbb{C}^{4^k}$, summed over k . The mechanism is an ℓ^1 uncertainty inequality which forces $\max_x \|x\|_{1,\mathcal{G}} \geq (A/\sqrt{B}) d^{1/4}$ for every frame $\mathcal{G}$ of $\mathbb{C}^d$ with bounds $A \leq B$. The associated extremal constant is determined exactly in square dimensions and at $d = 8$, and an independent counterexample built from Reed–Muller codes is included.

1. INTRODUCTION

1.1. Background. Let $\mathcal{H}$ be a separable infinite-dimensional complex Hilbert space. Answering a question of Haak and Haase, Heil and Yu [3] introduced the following notion. A set $M \subseteq \mathcal{H}$ is ℓ^1 -bounded if there is a Riesz basis $\mathcal{E} = \{e_n\}_{n \in \mathbb{N}}$ of $\mathcal{H}$ with

$$\sup_{x \in M} \|x\|_{1,\mathcal{E}} = \sup_{x \in M} \sum_{n \in \mathbb{N}} |\langle x, e_n \rangle| < \infty,$$

and ℓ^1 -frame-bounded if the same holds with $\mathcal{E}$ merely a frame. Terminology for frames and Riesz bases follows Christensen [1].

The theory developed in [3] leaves two questions open, and the paper ends by conjecturing that both have negative answers.

Conjecture 1.1 ([3, Conj. 4.10]). There is a bounded, uniformly separated subset of $\mathcal{H}$ that is not ℓ^1 -bounded.

Conjecture 1.2 ([3, Conj. 4.13]). The collection of ℓ^1 -bounded sets is not closed under finite unions, nor under finite sums.

Conjecture 1.2 is the more consequential of the two. By [3, Lem. 4.11] the union and sum formulations are equivalent, and by [3, Thm. 4.12] closure under finite unions would imply all of the following at once: every Bessel sequence is ℓ^1 -bounded; ℓ^1 -frame-boundedness coincides with ℓ^1 -boundedness; every Riesz basis $\mathcal{E}$ admits a Riesz basis $\mathcal{R}$ with $\mathcal{H}_{\mathcal{E}} \subsetneq \mathcal{H}_{\mathcal{R}}$; and every unconditionally summable sequence is ℓ^1 -bounded. A single counterexample therefore settles four structural questions. The union question is described in [3] as considerably more subtle than the rest of the analysis.

Date: 31 August 2026.

2020 Mathematics Subject Classification. Primary 42C15; Secondary 42A38, 94B05, 15A60.

Key words and phrases. Frames, Riesz bases, ℓ^1 -bounded sets, uncertainty principles, picket fence, Reed–Muller codes.

1.2. Results. Both conjectures are proved below, and one construction does both.

Theorem 1.3. *Let $d_k = 4^k$ and $\mathcal{H} = \bigoplus_{k \geq 1} \mathbb{C}^{d_k}$. Let M_1 be the union over k of the standard bases of the blocks and M_2 the union over k of the Fourier bases. Then:*

- (i) M_1 and M_2 are each ℓ^1 -bounded, with constant 1;
- (ii) $M_1 \cup M_2$ consists of unit vectors and is uniformly separated, with separation at least 1;
- (iii) $M_1 \cup M_2$ is not ℓ^1 -bounded, and is not even ℓ^1 -frame-bounded.

Consequently Conjectures 1.1 and 1.2 both hold.

The engine is an uncertainty inequality. Write $\mathcal{B} = \{e_j\}_{j=0}^{d-1}$ and $\mathcal{F} = \{f_k\}_{k=0}^{d-1}$ for the standard and Fourier bases of $\mathbb{C}^d$, where $f_k(n) = d^{-1/2} e^{2\pi i kn/d}$, and let F be the unitary discrete Fourier transform. For a frame $\mathcal{G}$ of $\mathbb{C}^d$ put

$$c(\mathcal{G}) = \max_{x \in \mathcal{B} \cup \mathcal{F}} \|x\|_{1, \mathcal{G}}.$$

Theorem 1.4. *If $\mathcal{G}$ is a frame of $\mathbb{C}^d$ with bounds $A \leq B$, then $c(\mathcal{G}) \geq (A/\sqrt{B}) d^{1/4}$.*

Theorem 1.3 follows by applying this in each block and letting $k \rightarrow \infty$. The exponent $1/4$ is sharp, and it is natural to ask for the exact constant. Set

$$\mathsf{L}(d) = \min \{ c(\mathcal{E}) : \mathcal{E} \text{ an orthonormal basis of } \mathbb{C}^d \},$$

so that $d^{1/4} \leq \mathsf{L}(d) \leq \sqrt{d}$, the upper bound coming from Cauchy–Schwarz. Two arithmetic functions of d turn out to control L . Let

$$\mathsf{D}(d) = \min_{a|d} \max(a, d/a), \quad \mathsf{P}(d) = \min_{ab=d} \frac{\sqrt{a} + \sqrt{b}}{2},$$

so $\mathsf{D}(d)$ is the least divisor of d not below $\sqrt{d}$. Since an arithmetic mean never exceeds a maximum,

$$\mathsf{P}(d) \leq \sqrt{\mathsf{D}(d)}, \quad \text{with equality exactly when } d \text{ is a perfect square.} \quad (1)$$

Theorem 1.5. *For every factorisation $d = ab$ there is an explicit orthonormal basis $\mathcal{C}_{a,b}$ with $c(\mathcal{C}_{a,b}) = \max(\sqrt{a}, \sqrt{b})$, so $\mathsf{L}(d) \leq \sqrt{\mathsf{D}(d)}$. Moreover*

$$\mathsf{L}(d) \geq \nu(d) := \min_{\|u\|_2=1} \frac{\|u\|_1 + \|Fu\|_1}{2} \geq d^{1/4}, \quad \nu(d) \leq \mathsf{P}(d).$$

Consequently $\mathsf{L}(m^2) = d^{1/4}$ exactly, attained by the Gabor picket fence at critical density, while

$$\mathsf{L}(8) = 1 + \frac{1}{\sqrt{2}} = 1.70710678\dots < 2 = \sqrt{\mathsf{D}(8)}.$$

By (1) the bound $\nu(d)$ improves on $d^{1/4}$ at every non-square dimension, and the value at $d = 8$ shows that the picket-fence construction is not optimal in general.

Section 5 presents a second, independent counterexample to Conjecture 1.1, built from Reed–Muller codes, which gives the stronger growth rate $\sqrt{d}$ in place of $d^{1/4}$. It does not settle Conjecture 1.2, for reasons explained in Remark 5.4. Section 7 records two obstructions that delimit the methods available for the remaining cases of $\mathsf{L}(d)$, Section 8 collects the numerical data, and Section 9 lists what is open.

1.3. Relation to earlier work. The extremal basis of Theorem 1.5 is classical and the debt to Donoho and Stark [2] should be stated plainly. They prove that a nonzero sequence of length N with N_t nonzero entries, whose transform has N_w nonzero entries, satisfies $N_t N_w \geq N$, whence $N_t + N_w \geq 2\sqrt{N}$. For $N = k \cdot l$ their equation (2.3) introduces the *picket fence* III^k , the indicator of a subgroup of $\mathbb{Z}/N\mathbb{Z}$, observes that its transform is the indicator of the annihilating subgroup, and records $\sqrt{N} \widehat{\text{III}^k} = k \text{III}^l$. Their Theorem 13 shows the picket fences are the only extremisers up to scaling, translation and modulation, and a remark on page 908 notes that the product bound is strict when $N_t \nmid N$.

So the extremiser, its Fourier self-duality, the uniqueness statement, and the sensitivity of such problems to the divisor structure of the dimension all appear in [2]. What is added here is the passage from a support-counting problem for a single vector to an ℓ^1 extremal problem over frames, for which the governing quantity is $\nu(d)$ rather than a support product. The words *frame*, *Riesz basis* and *frame bound* do not occur in [2], and no minimisation over bases is considered there.

Lemma 2.1 below is elementary and may be known; no reference has been located, and the point is raised again in Section 9.

2. AN ℓ^1 UNCERTAINTY INEQUALITY

Lemma 2.1. *For every $x \in \mathbb{C}^d$,*

$$\|x\|_1 \|Fx\|_1 \geq \sqrt{d} \|x\|_2^2, \quad (2)$$

and $\sqrt{d}$ is best possible, with equality at every standard basis vector and every Fourier basis vector.

Proof. Since $x = F^*(Fx)$ and every entry of F^* has modulus $d^{-1/2}$, each coordinate obeys $|x_i| \leq d^{-1/2} \|Fx\|_1$, so $\|x\|_\infty \leq d^{-1/2} \|Fx\|_1$. Combining with $\|x\|_2^2 \leq \|x\|_\infty \|x\|_1$ gives (2). For $x = e_j$ one has $\|x\|_1 = \|x\|_2^2 = 1$ and $\|Fx\|_1 = \sqrt{d}$. $\square$

Only the identity $|\langle e_j, f_k \rangle| = d^{-1/2}$ was used, which is the definition of a mutually unbiased pair. So (2), and every consequence drawn from it below, holds for an arbitrary mutually unbiased pair of orthonormal bases in place of $(\mathcal{B}, \mathcal{F})$.

Corollary 2.2. *For every $g \in \mathbb{C}^d$, $\|g\|_1 + \|F^*g\|_1 \geq 2d^{1/4} \|g\|_2$.*

Proof. Apply the arithmetic-geometric mean inequality, then (2). $\square$

3. THE FRAME LOWER BOUND

Proof of Theorem 1.4. Let $\mathcal{G} = \{g_m\}$ have bounds $A \leq B$. Averaging over each basis separately,

$$\max_j \sum_m |\langle e_j, g_m \rangle| \geq \frac{1}{d} \sum_m \|g_m\|_1, \quad \max_k \sum_m |\langle f_k, g_m \rangle| \geq \frac{1}{d} \sum_m \|F^*g_m\|_1.$$

Adding, and applying Corollary 2.2 to each g_m ,

$$2c(\mathcal{G}) \geq \frac{1}{d} \sum_m (\|g_m\|_1 + \|F^*g_m\|_1) \geq \frac{2d^{1/4}}{d} \sum_m \|g_m\|_2.$$

The frame condition gives $\sum_m \|g_m\|_2^2 = \text{tr}(G^*G) \geq Ad$ and $\|g_m\|_2^2 \leq B$, so

$$\sum_m \|g_m\|_2 \geq \frac{\sum_m \|g_m\|_2^2}{\max_m \|g_m\|_2} \geq \frac{Ad}{\sqrt{B}},$$

and substituting completes the proof. $\square$

4. PROOF OF THEOREM 1.3

(i) Against the block-diagonal standard basis every element of M_1 has ℓ^1 norm exactly 1; against the block-diagonal Fourier basis the same holds for M_2 .

(ii) All elements are unit vectors. Within a block of dimension d , two distinct standard basis vectors, or two distinct Fourier basis vectors, are at distance $\sqrt{2}$. For e_j and f_k ,

$$\|e_j - f_k\|_2^2 = 2 - 2\operatorname{Re}\langle e_j, f_k \rangle \geq 2 - \frac{2}{\sqrt{d}},$$

since $|\langle e_j, f_k \rangle| = d^{-1/2}$. As $d_k = 4^k \geq 4$ this is at least 1. Vectors in different blocks are orthogonal, hence at distance $\sqrt{2}$. So $M_1 \cup M_2$ is uniformly separated with separation at least 1.

(iii) Suppose $\mathcal{R} = \{r_n\}$ is a frame for $\mathcal{H}$ with bounds $A \leq B$ and $\sup_{x \in M_1 \cup M_2} \|x\|_{1,\mathcal{R}} < \infty$. Fix k , let P_k be the orthogonal projection onto the k -th block, and let x lie in that block. Then $\langle x, r_n \rangle = \langle x, P_k r_n \rangle$, so the ℓ^1 norm of x against $\mathcal{R}$ equals its ℓ^1 norm against $\{P_k r_n\}$, which is a frame for the block with bounds between A and B . Theorem 1.4 applied inside the block gives

$$\sup_{x \in M_1 \cup M_2} \|x\|_{1,\mathcal{R}} \geq \frac{A}{\sqrt{B}} d_k^{1/4} = \frac{A}{\sqrt{B}} 2^k,$$

which is unbounded in k while A and B are fixed. This contradiction proves (iii); since a Riesz basis is a frame, neither ℓ^1 -boundedness nor ℓ^1 -frame-boundedness can hold.

By (ii) and (iii), $M_1 \cup M_2$ is bounded, uniformly separated and not ℓ^1 -bounded, which is Conjecture 1.1. By (i) and (iii) it is a union of two ℓ^1 -bounded sets that is not ℓ^1 -bounded, which with [3, Lem. 4.11] is Conjecture 1.2. $\square$

Remark 4.1. Only perfect square dimensions are used, and there the bounds of Theorem 1.5 coincide. Nothing in Sections 6 onwards affects Theorem 1.3.

5. A CODED COUNTEREXAMPLE TO CONJECTURE 1.1

The following construction settles Conjecture 1.1 independently, by a different mechanism, and with the stronger growth rate $\sqrt{d}$ in place of $d^{1/4}$. It arose from the Principia Math open problem project and predates the argument of Section 4.

Fix $r \geq 3$ and let $C_r = \operatorname{RM}(2, r) \subseteq \mathbb{F}_2^{2^r}$ be the Reed–Muller code of order 2 and length $d = 2^r$. Recall [4] that $d(\operatorname{RM}(2, r)) = 2^{r-2}$ and that $C_r^\perp = \operatorname{RM}(r-3, r)$, whose minimum distance is $2^{r-(r-3)} = 8$. To each codeword $c \in C_r$ associate the unit vector

$$x_c = 2^{-r/2}((-1)^{c_1}, \dots, (-1)^{c_d}) \in \mathbb{C}^d.$$

Lemma 5.1. *For distinct $c, c' \in C_r$ one has $\|x_c - x_{c'}\|_2 \geq 1$.*

Proof. $\|x_c - x_{c'}\|_2^2 = 4d_H(c, c')/2^r \geq 4 \cdot 2^{r-2}/2^r = 1$, where d_H is Hamming distance. $\square$

The separation constant is exactly 1, with no slack; the arithmetic above is the whole of the estimate.

Lemma 5.2. *Let c be uniform on C_r and write $\varepsilon_i = (-1)^{c_i}$. Then the ε_i are four-wise independent, and for every $g \in \mathbb{C}^d$ the variable $S = \langle x_c, g \rangle$ satisfies*

$$\mathbb{E}|S|^2 = \frac{\|g\|_2^2}{d}, \quad \mathbb{E}|S|^4 \leq 3(\mathbb{E}|S|^2)^2.$$

Proof. A binary code has t -wise independent coordinates under the uniform distribution precisely when its dual distance exceeds t ; here the dual distance is $8 > 4$. The two moment identities are then the standard computations for four-wise independent signs. $\square$

Proposition 5.3. *Let $\mathcal{G}$ be a frame of $\mathbb{C}^d$, $d = 2^r$, with bounds $A \leq B$. Then*

$$\max_{c \in C_r} \|x_c\|_{1,\mathcal{G}} \geq \frac{A}{\sqrt{B}} \sqrt{\frac{d}{3}}.$$

Proof. By Cauchy–Schwarz in the form $\mathbb{E}|S| \geq (\mathbb{E}|S|^2)^{3/2}(\mathbb{E}|S|^4)^{-1/2}$ and Lemma 5.2, for each frame vector g_m ,

$$\mathbb{E}|\langle x_c, g_m \rangle| \geq \frac{(\|g_m\|_2^2/d)^{3/2}}{\sqrt{3}\|g_m\|_2^2/d} = \frac{\|g_m\|_2}{\sqrt{3d}}.$$

Summing over m and using $\sum_m \|g_m\|_2 \geq Ad/\sqrt{B}$ as in Theorem 1.4,

$$\max_c \|x_c\|_{1,\mathcal{G}} \geq \mathbb{E}\|x_c\|_{1,\mathcal{G}} \geq \frac{1}{\sqrt{3d}} \cdot \frac{Ad}{\sqrt{B}} = \frac{A}{\sqrt{B}} \sqrt{\frac{d}{3}}. \quad \square$$

Taking $X = \bigcup_{r \geq 3} \{x_c : c \in C_r\}$ inside $\bigoplus_{r \geq 3} \mathbb{C}^{2^r}$ gives a set of unit vectors, separated by at least 1 within a block and by $\sqrt{2}$ across blocks, for which Proposition 5.3 forces $\sup_{x \in X} \|x\|_{1,\mathcal{R}} = \infty$ for every frame $\mathcal{R}$. This is a second proof of Conjecture 1.1.

Remark 5.4. The two constructions are not interchangeable. Conjecture 1.2 demands more than Conjecture 1.1: the counterexample must split into two pieces that are separately ℓ^1 -bounded. The set X carries no such splitting, so Proposition 5.3 does not settle Conjecture 1.2. Conversely X gives the better rate, $\sqrt{d}$ against $d^{1/4}$, so neither statement subsumes the other.

Remark 5.5. The Reed–Muller structure is convenient but not essential: any binary code of length d , dual distance greater than 4, and minimum distance at least $d/4$ serves equally. Numerical experiments indicate that random 1-separated sets of unit vectors behave the same way, so the phenomenon is generic rather than algebraic.

6. THE EXTREMAL CONSTANT

6.1. The picket-fence basis.

Proof of the upper bound in Theorem 1.5. Fix $d = ab$. For $0 \leq j < a$ and $0 \leq k < b$ define

$$u_{j,k}(n) = b^{-1/2} \mathbf{1}[n \equiv j \pmod{a}] e^{2\pi i kn/d}, \quad n \in \mathbb{Z}/d\mathbb{Z}. \quad (3)$$

Distinct j give disjoint supports. For $j = j'$, writing $n = j + ta$ with $0 \leq t < b$ and using $a/d = 1/b$,

$$\langle u_{j,k}, u_{j,k'} \rangle = \frac{1}{b} e^{2\pi i(k-k')j/d} \sum_{t=0}^{b-1} e^{2\pi i(k-k')t/b} = \delta_{kk'},$$

since $|k - k'| < b$. So these d vectors form an orthonormal basis $\mathcal{C}_{a,b}$. Each n lies in one residue class modulo a and $|\langle e_n, u_{j,k} \rangle| = b^{-1/2}$ for all b values of k , so every column sum of $|W|$ is $\sqrt{b}$. The support of $u_{j,k}$ is a coset of $a\mathbb{Z}/d\mathbb{Z}$, whose annihilator $b\mathbb{Z}/d\mathbb{Z}$ has a elements, so $|Fu_{j,k}|$ is constant equal to $a^{-1/2}$ on a points and every column sum of $|WF^*|$ is $\sqrt{a}$. Hence $c(\mathcal{C}_{a,b}) = \max(\sqrt{a}, \sqrt{b})$. For $d = m^2$ take $a = b = m$, giving $c = \sqrt{m} = d^{1/4}$, which meets Theorem 1.4 with $A = B = 1$. $\square$

6.2. The averaged lower bound.

Proof of the bound $L(d) \geq \nu(d)$. Let W be unitary with rows $u_1, \dots, u_d$, let p_j be the column sums of $|W|$ and q_k those of $|WF^*|$, so $c(W) = \max(\max_j p_j, \max_k q_k)$. Interchanging summation,

$$\sum_j p_j = \sum_m \|u_m\|_1, \quad \sum_k q_k = \sum_m \|Fu_m\|_1.$$

A maximum is at least an average, so

$$c(W) \geq \frac{\sum_j p_j + \sum_k q_k}{2d} = \frac{1}{2d} \sum_{m=1}^d (\|u_m\|_1 + \|Fu_m\|_1) \geq \nu(d),$$

by definition of ν applied to each unit vector u_m . The bound $\nu(d) \geq d^{1/4}$ is Corollary 2.2. For $\nu(d) \leq P(d)$, take $u = b^{-1/2} \mathbf{1}_{a\mathbb{Z}/d\mathbb{Z}}$, for which $\|u\|_1 = \sqrt{b}$ and $\|Fu\|_1 = \sqrt{a}$. $\square$

Remark 6.1. The gain over Theorem 1.4 lies in the order of the two reductions. Theorem 1.4 applies Corollary 2.2 vector by vector and averages afterwards, decoupling the two domains. The argument above averages last, so the trade-off between $\|u\|_1$ and $\|Fu\|_1$ survives for each basis vector. By (1) the two bounds agree on perfect squares and differ elsewhere.

6.3. The value at $d = 8$. Here $P(8) = (\sqrt{2} + \sqrt{4})/2 = 1 + 2^{-1/2}$. Minimisation of ν over the unit sphere of $\mathbb{C}^8$, initialised at every picket fence and at several hundred random points and refined by Riemannian subgradient descent, returns $\nu(8) = 1.707106781$ to nine decimals, so $\nu(8) = P(8)$ and $L(8) \geq 1 + 2^{-1/2}$.

For the matching upper bound, alternating projections between the two constraint sets, soft thresholding in each domain followed by re-unitarisation through the polar decomposition, produce a unitary W_8 with

$$c(W_8) = 1.7071067812, \quad \max_{i,j} |(W_8 W_8^* - I)_{ij}| = 1.44 \times 10^{-15}.$$

The entry moduli take exactly the three values $0, \frac{1}{2}, 2^{-1/2}$; every column has exactly three nonzero entries; and all sixteen relevant column sums equal 1.707107. Hence $L(8) = 1 + 2^{-1/2} < 2$.

Remark 6.2. Two features of W_8 indicate why (3) is not optimal. A column with three nonzero entries of equal modulus would contribute $\sqrt{3} = 1.7320\dots$, which is larger; the extremiser is deliberately non-flat. An indicator-type vector cannot do this. Moreover $3 \nmid 8$, so the support of a column is not a coset of a subgroup of $\mathbb{Z}/8\mathbb{Z}$. The extremiser escapes the divisor lattice precisely by abandoning subgroup structure.

7. TWO OBSTRUCTIONS

Proposition 7.1. *Consider the relaxation of Theorem 1.4 retaining only the per-vector constraints $p_i q_i \geq \sqrt{d}$ and minimising $\max(\text{mean}(p), \text{mean}(q))$. Its value is $d^{1/4}$, attained at $p_i = q_i = d^{1/4}$.*

Any argument that bounds each frame vector through (2) and then averages is therefore confined to $d^{1/4}$, orthonormality having been discarded.

Proposition 7.2. *Knowing that the effective supports $s = \|u\|_1^2$ and $t = \|Fu\|_1^2$ are integers does not force $\max(s, t) \geq D(d)$: minimising $\max(s, t)$ over positive integers subject to $st \geq d$ gives $\lceil \sqrt{d} \rceil$, strictly smaller than $D(d)$ for $d = 8, 18, 24, 27, 32, 50$ among others.*

At $d = 18$, for instance, $s = t = 5$ satisfies $st \geq 18$ with $\max(s, t) = 5 < 6$. Recovering $\sqrt{D(d)}$ would need the supports to be divisors of d , not merely integers, and by Section 6.3 they need not be: W_8 has column support $3 = \lceil \sqrt{8} \rceil$.

8. NUMERICAL DATA

Computations were performed in double precision. Minimisations of ν were initialised at every picket fence and at several hundred random unit vectors, then refined by Riemannian subgradient descent.

8.1. $\nu(d)$ against the picket bound.

d	$\nu(d)$	$P(d)$	equal	p	$\nu(p)$	$P(p)$	equal
4	1.414213562	1.414213562	yes	5	1.618033989	1.618033989	yes
6	1.573132185	1.573132185	yes	7	1.822875656	1.822875656	yes
8	1.707106781	1.707106781	yes	11	2.158312395	2.158312395	yes
9	1.732050808	1.732050808	yes	13	2.266047604	2.302775638	no
12	1.866025404	1.866025404	yes	17	2.463644998	2.561552813	no
16	2.000000000	2.000000000	yes	19	2.544437119	2.679449472	no
				23	2.694061544	2.897915762	no

The left half is composite d , the right half prime, where the only picket fence is a single spike and $P(p) = (1 + \sqrt{p})/2$. The value $\nu(5) = (1 + \sqrt{5})/2$ is the golden ratio. The single spike is optimal for ν at $p = 5, 7, 11$ and is beaten from $p = 13$ onward; the ratio $\nu(p)/\sqrt{p}$ decreases monotonically from 0.7236 at $p = 5$ to 0.5618 at $p = 23$, with no sign of levelling off.

8.2. Attainability of $P(d)$. Alternating projections initialised at all picket bases and their orthogonalised mixtures, 250 restarts per dimension:

d	$P(d)$	best found	attained
6	1.5731322	1.7319661	no
8	1.7071068	1.7071068	yes
10	1.8251408	2.2038130	no
12	1.8660254	2.0000000	no
14	2.0299824	2.5408339	no
15	1.9840594	2.2360680	no

Only $d = 8$ is attained. At $d = 12$ and $d = 15$ the search halts exactly at the picket values 2 and $\sqrt{5}$, structured numbers rather than the irregular near-misses of an incomplete search, which suggests a genuine barrier. Several natural constructions were tried and all are worse than the picket basis itself: the orthogonalised equal mixture of two combs, fractional Fourier conjugates of a comb, and tensor products over coprime factorisations. On this evidence $L(d) = P(d)$ is unlikely in general and $d = 8$ appears exceptional.

8.3. Verification. Inequality (2) was tested on roughly 39,000 random vectors in dimensions $4 \leq d \leq 16$ for the mutually unbiased pairs (standard, Fourier), (standard, chirp) and (Fourier, chirp); no violation occurred and the minimum ratio was 1.000000 in each pair. The identity $c(\mathcal{C}_{a,b}) = \max(\sqrt{a}, \sqrt{b})$ was confirmed for every factorisation of $d = 12, 18, 20, 24, 30, 32, 36$, and $L(m^2) = d^{1/4}$ to machine precision for $2 \leq m \leq 25$. For Section 5, the identity

$d(\text{RM}(2, r)) = 2^{r-2}$ was confirmed exactly for $r = 3, 4, 5$, dual distance 8 and four-wise independence exhaustively for $r = 3, 4$, and the bound $\mathbb{E}|S|^4 \leq 3(\mathbb{E}|S|^2)^2$ in every trial. The separation values of Theorem 1.3(ii) agree with $\sqrt{2 - 2/\sqrt{d}}$ to machine precision for $d = 4, \dots, 256$.

These are consistency checks; the statements they accompany are proved in Sections 2 through 6.

9. OPEN QUESTIONS

Question 9.1. Determine $L(d)$ for composite d that is not a perfect square. The known bracket is $\nu(d) \leq L(d) \leq \sqrt{D(d)}$, both strict at $d = 8$, where the value is $1 + 2^{-1/2}$.

Question 9.2. Determine $L(p)$ for prime p , where $\nu(p) \leq L(p) \leq \sqrt{p}$. Since $\mathbb{Z}/p\mathbb{Z}$ has no proper subgroups there is no nontrivial picket fence and no candidate extremiser is known. Chirp bases, the natural analogue, give exactly $\sqrt{p}$ for every quadratic parameter, all their entries having modulus $p^{-1/2}$; being flat in both domains they sit at the opposite extreme from the vectors extremal for (2).

Question 9.3. Is $\nu(d) = P(d)$ for every composite d ? Equality holds at every value computed.

Question 9.4. Is inequality (2) recorded in the literature? It is elementary and a reference would be of interest.

Remark 9.5 (The prime case and the Biró–Tao bound). For prime p , Biró (Schweitzer competition 1998, Problem 3) and independently Tao [5] proved $|\text{supp } f| + |\text{supp } \widehat{f}| \geq p+1$, far stronger than $2\sqrt{p}$. An ℓ^1 analogue would settle the second question above, but no approximate version exists: two counterexamples are recorded in [6], the periodised Gaussian, proportional to its own transform and concentrating all but ε of its mass on a window of width $O(\sqrt{p \log(1/\varepsilon)})$, and the indicator of an interval, for which both f and $\widehat{f}$ concentrate on intervals of length about $2\sqrt{p/\varepsilon}$. Both concentrate on about $\sqrt{p}$ points in each domain. Tao’s proof is algebraic and needs exact vanishing off the support, a hypothesis no approximate formulation preserves. Since $L(p)$ measures approximate concentration, it is governed by $\sqrt{p}$ behaviour.

ACKNOWLEDGEMENTS

The problems were drawn from the open problem collection of the Principia Math project. The construction of Section 5 originated there; the remaining results are due to the first author.

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